

ПРАВИТЕЛЬСТВО РОССИЙСКОЙ ФЕДЕРАЦИИ
ФГАОУ ВО НАЦИОНАЛЬНЫЙ ИССЛЕДОВАТЕЛЬСКИЙ УНИВЕРСИТЕТ
«ВЫСШАЯ ШКОЛА ЭКОНОМИКИ»

Факультет компьютерных наук
Образовательная программа «Прикладная математика и информатика»

УДК 004.89

Отчет об исследовательском проекте на тему:
Музыкальная транскрипция гитары по аудиовизуальному сигналу
(промежуточный, этап 1)

Выполнил студент:

группы #БПМИ233, 3 курса

Сергеев Николай Александрович

Принял руководитель проекта:

Гринберг Петр Маркович

Научный сотрудник

Факультет компьютерных наук НИУ ВШЭ

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Annotation

Automatic Music Transcription (AMT) refers to the task of converting an audio recording of an instrumental performance into a symbolic representation such as sheet music, MIDI, or tablature. Although many guitar pieces have already been transcribed into sheet music or tablature, a large number of complex musical performances remain undocumented.

For most guitarists, manual transcription by ear is an extremely tedious, time-consuming, and often infeasible process. Consequently, robust and accurate automatic guitar transcription systems would significantly lower the barrier to music learning and enable broader access to musical education. Furthermore, such methods could serve as the foundation for affordable, fully automated platforms capable of teaching guitar and providing precise, real-time performance assessment.

Despite the guitar’s widespread popularity, AMT for this instrument has received considerably less attention than piano transcription. Existing approaches often under-perform on fast or polyphonic passages and frequently fail to recover correct fretting positions. Moreover, the very definition of an “optimal” guitar transcription must be carefully reconsidered in light of the instrument’s unique mechanics and the scarcity of high-quality annotated datasets.

The primary objective of this planned work is threefold: (1) to survey existing AMT methods and datasets tailored to the guitar, (2) to identify their key limitations, and (3) to propose a novel deep neural network (DNN) architecture that leverages audio-visual signals to overcome these challenges—particularly in recovering accurate string–fret assignments and supporting alternative tunings.

Аннотация

Автоматическая транскрипция музыки — задача преобразования аудиозаписи инструментального исполнения в символическое представление, такое как нотная запись или табулатура. Хотя многие гитарные произведения уже переведены в ноты или табулатуру, значительное число сложных музыкальных пьес остаётся не задокументированным.

Для большинства гитаристов ручная транскрипция «на слух» является крайне утомительной, трудоёмкой и зачастую невозможной задачей. Следовательно, надёжные и точные системы автоматической транскрипции гитарной музыки значительно снизили бы порог входа в музыкальное обучение и расширили бы доступ к музыкальному образованию. Более того, такие методы могли бы лечь в основу доступных полностью автоматизированных

платформ, способных обучать игре на гитаре и обеспечивать точную оценку исполнения в реальном времени.

Несмотря на широкую популярность гитары, автоматическая транскрипция для этого инструмента получила значительно меньше внимания по сравнению с транскрипцией фортепианной музыки. Существующие подходы часто демонстрируют недостаточную эффективность при быстрых или многоголосных пассажах и нередко ошибаются в определении позиций ладов. Кроме того, само понятие «оптимальной» гитарной транскрипции требует тщательного переосмысления с учётом особенностей устройства инструмента и дефицита качественно размеченных наборов данных.

Цель данной работы: (1) провести обзор существующих методов автоматической транскрипции гитары и наборов данных; (2) выявить их ключевые ограничения; (3) предложить новую архитектуру глубокой нейронной сети, использующую совместный анализ звукового и видеосигнала для преодоления выявленных трудностей — в первую очередь для точного восстановления соответствия между струнами и ладами, а также для поддержки альтернативных строев.

Keywords

Automatic Music Transcription, Guitar Tablature, Multimodal Learning, String-Fret Ambiguity, Alternative Tunings, Onset Detection, Pitch Regression, Audio-Visual Fusion.

1 Introduction

The guitar is a highly expressive yet mechanically complex instrument. Guitarists employ a wide range of techniques—including hammer-ons, pull-offs, tapping, and natural or artificial harmonics—that involve intricate interactions between finger placement, string excitation, and timing. These nuances pose significant challenges for automatic transcription systems.

Historically, guitar music has been notated using two primary formats: standard sheet music and tablature (tabs). Sheet music provides a universal, standardized representation that encodes both pitch and rhythm. However, it does not specify how a note should be played on the guitar (i.e., which string and fret to use). In contrast, tablature explicitly indicates string and fret positions, making it far more intuitive and practical for guitarists. As a result, most guitar-focused AMT systems aim to generate tablature rather than conventional notation.

As illustrated in Figure 1.1, guitar tabs are commonly represented in a simple ASCII format. Notably, such representations often omit explicit rhythmic information. While this is generally acceptable to experienced players—who can infer timing from context—the core objective of guitar transcription remains the accurate identification of which string and fret produce each note.

The figure displays three musical representations of J.S. Bach's Bourree, organized into two columns. The left column contains the first measure, and the right column contains the second measure. Above the columns is a section labeled 'B II' with a double bar line and a repeat sign.

- Standard notation:** The top staff shows musical notes with stems and flags. Fingerings are indicated by letters 'i' (index) and 'm' (middle). The first measure is marked with a '1' and a repeat sign. The second measure is marked with a '2'.
- Tablature:** The middle staff shows fret numbers (0, 2, 3, 4) placed on a six-line staff. The first measure is marked with a '1' and a repeat sign. The second measure is marked with a '2'.
- ASCII Tablature:** The bottom staff shows fret numbers (0, 2, 3, 4) placed on a six-line staff. The first measure is marked with a '1' and a repeat sign. The second measure is marked with a '2'.

Figure 1.1: Standard notation, Tablature, ASCII Tablature comparison. J.S. Bach Bourree.

That said, more informative tablature would ideally include temporal annotations. In practice, however, precise note duration modeling is less critical for the guitar than for instruments

like the piano. The guitar typically has only six strings, and notes are often terminated by the onset of subsequent notes due to limited sustain and the physical necessity of repositioning fingers quickly. Moreover, guitarists frequently allow notes to ring as long as musically appropriate, meaning that actual note durations are highly dependent on individual playing style and artistic interpretation—not just the underlying score. Thus, transcription systems that omit exact offset times can still yield musically useful results.

Rhythmic expression further complicates the task. As noted in [9], certain stylistic elements—such as swing feel—are rarely encoded in standard notation but are essential to authentic performance. Similarly, dynamics, articulation, and timbral variations are often conveyed through marginal annotations or performer intuition rather than explicit symbols. While the cited work addresses the inverse problem (generating expressive performances from scores), it highlights a shared challenge between music generation and AMT: the gap between symbolic notation and acoustic reality.

For guitar AMT, the highest priority is therefore the reliable and efficient recovery of essential musical content: namely, pitch, string, and fret. Secondary attributes such as dynamics or precise note offsets may be addressed in future work.

A fundamental challenge unique to stringed instruments is positional ambiguity: a single pitch can often be played on multiple strings and at different frets (see Figure 1.2). Consequently, even a correctly identified pitch may correspond to several valid tablature representations. This ambiguity is exacerbated by advanced playing techniques and, critically, by non-standard tunings. Many classical and contemporary pieces employ alternate tunings (e.g., Drop D), and modern guitarists frequently experiment with custom configurations. An ideal transcription system should be robust to such variations without requiring retraining or manual intervention.

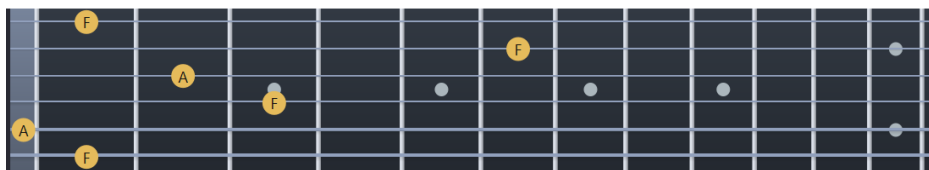


Figure 1.2: Guitar fretboard notes example.

2 Related Work

Given the unique challenges of guitar tablature transcription (GTT) several approaches have been proposed. Early methods relied on traditional audio signal processing [5] or probabilistic

modeling [13]. More recently, deep neural networks have emerged as the dominant paradigm due to their ability to learn complex mappings from audio to symbolic representations.

Among these, TabCNN [11] is widely regarded as the state-of-the-art for GTT. Unlike generic pitch transcription systems, TabCNN is explicitly designed to predict fingering positions—i.e., string–fret pairs—enabling direct generation of guitar tablature. The model processes Constant-Q Transform (CQT) spectrograms, which provide a musically meaningful frequency representation by aligning bins with musical pitches. Each time frame is treated as an image-like input to a convolutional neural network (CNN), producing a 6×21 output matrix representing softmax probabilities over frets for each of the six strings. The authors also introduce evaluation metrics tailored to GTT, acknowledging that while their system operates at the frame level, its outputs can be post-processed into note-level transcriptions.

Subsequent work has shifted toward note-level and tokenized representations. Kim et al. [7] extended the TabCNN architecture by incorporating self-attention mechanisms and beat-aligned temporal context, enabling direct estimation of discrete note events. This approach demonstrated competitive performance not only in note-level transcription but also in frame-level metrics. In a follow-up study [8], the same authors adopted a tokenized output format, inspired by modern automatic speech recognition (ASR). Their sequence-to-sequence Transformer model employs Connectionist Temporal Classification (CTC) loss to handle unaligned input–output sequences—a technique particularly valuable given the scarcity of precisely annotated guitar datasets. However, because the target representation is based on MIDI, which encodes pitch and timing but not string or fret information, the original fingering is irreversibly lost.

To address this limitation, recent work has explored fingering recovery from MIDI. The Fretting Transformer [4] uses an attention-based model to map MIDI notes to plausible string–fret assignments. While it produces musically valid tablature, it cannot guarantee fidelity to the performer’s original fingering, especially in passages with multiple valid fingerings.

It is worth noting that nearly all existing guitar transcription systems—including TabCNN and even early token-based adaptations—ultimately rely on an intermediate or final frame-level piano-roll representation, where the model predicts, for every time frame, which pitches are active. While this formulation aligns naturally with spectrogram inputs and enables the use of standard losses like binary cross-entropy, it suffers from several fundamental drawbacks. First, it requires precise frame-wise annotations of note onsets and, more problematically, offsets—yet as discussed earlier, offset timing in guitar performance is often ambiguous, stylistic, and inconsistent across players. This introduces significant label noise and increases annotation cost. Second, the piano-roll format conflates acoustic presence with symbolic intent: a ringing note may be acoustically

present long after its notated duration, leading to mismatches between ground truth and performer interpretation.

Nevertheless, the most common limitation across all aforementioned methods is their reliance on audio-only inputs. Although audio provides reliable pitch information, it offers little direct evidence about which string produced a given note—a fundamental ambiguity in guitar transcription. In contrast, visual signals contain explicit spatial cues: the plucking hand reveals string activation order, while the fretting hand indicates precise finger positions. Despite this, few AMT systems have leveraged video. Notably, Asmar [1] investigated vision-only GTT using RGB-D video and classical computer vision techniques—including object segmentation, optical flow, Hough transforms, and clustering. While the system underperformed compared to audio-based methods, it demonstrated that visual data alone can support meaningful transcription, particularly when audio is unavailable or degraded.

More recently, multimodal approaches combining audio and video have gained traction. TapToTab [3] proposes a pipeline that uses YOLO to detect and segment the fretboard in video frames, then infers active fret regions via visual analysis, which are fused with audio-derived pitch estimates. The authors report improved fret prediction accuracy, underscoring the complementary nature of the two modalities.

A persistent bottleneck across all deep learning-based GTT systems is the lack of large-scale, high-quality datasets. Public audio datasets such as GuitarSet [12], AGPT [2], and IDMT-SMT-Guitar [6] exist, but many researchers consider them insufficient for training modern architectures without extensive data augmentation or synthetic pretraining. Synthetic audio—often generated from MIDI using simple oscillators—is commonly used to scale training, though it fails to capture realistic timbral and expressive nuances.

Critically, no public video datasets for GTT are currently available. This gap stems from the labor-intensive nature of annotation: accurate tablature requires precise labeling of both onsets (when a note begins) and fret-string assignments, often necessitating specialized hardware (e.g., hexaphonic pickups) or expert manual review. Moreover, as noted earlier, offset timing is inherently ambiguous in guitar performance and may introduce noise if included rigidly in labels.

These challenges motivate our work: we seek to develop a data-efficient, multimodal GTT framework that (1) explicitly recovers string-fret assignments using visual cues, (2) minimizes reliance on precisely aligned onset/offset annotations, and (3) generalizes across alternative tunings—thereby addressing core limitations of existing approaches.

3 Future Plans

Following the paradigm established by TabCNN, we adopt a frame-level formulation wherein each time step (e.g., 10 ms) yields predictions per string. However, unlike TabCNN—which performs multi-class classification over frets—we decouple pitch estimation from physical fingering. Specifically, for each of the six strings, the model will predict: A binary onset activation (indicating whether the string was plucked at that frame), A continuous MIDI pitch value (representing the sounding note). This design choice enables explicit modeling of note onsets—a prerequisite for accurate temporal alignment—while avoiding premature commitment to discrete fret positions. Onset detection will leverage spectrogram inputs processed by convolutional neural networks (CNNs), a strategy validated in prior work on musical onset detection [10].

A critical limitation of classification-based approaches (e.g., predicting fret numbers 0–24) is their rigidity under alternative tunings. By regressing MIDI pitch values instead, our model becomes inherently tuning-agnostic: given a known tuning configuration as side information, the fret number can be derived post-hoc via simple arithmetic. Moreover, MIDI pitch provides a linearly spaced representation of log-frequency, making it more suitable for regression than raw frequency (Hz). To reflect the perceptual similarity between adjacent frets, we employ a regression loss (e.g., mean squared error) rather than categorical cross-entropy. The total loss function combines: Binary cross-entropy for string-wise onset detection, Mean squared error for MIDI pitch regression, applied only to frames where an onset is active. This selective application prevents the model from being penalized for pitch errors on silent strings.

While CNNs provide strong local feature extraction, they may lack sufficient temporal context for disambiguating rapid passages or sustained notes. To address this, we consider augmenting the CNN backbone with temporal modeling modules, such as: Long Short-Term Memory (LSTM) layers or Temporal Convolutional Networks (TCNs). These additions will allow the model to integrate contextual information across neighboring frames, improving robustness to tempo variations and polyphonic interference. To further resolve ambiguities inherent in audio-only processing, we propose integrating close-up video footage of the guitar’s soundhole region (near the picking hand). This view maximizes visibility of string vibrations while minimizing occlusion from the fretting hand. We plan to process video frames using a TCN-based encoder to capture fine-grained string motion patterns indicative of plucking events. An even more refined approach would involve explicit string segmentation (e.g., via U-Net or keypoint detection), which could isolate individual string activity and suppress noise from hand movements. Crucially, audio and video streams must be temporally synchronized at the frame level. Given the absence of publicly available datasets

containing aligned audio–video recordings with precise onset and string annotations, we recognize the necessity of curating a new multimodal dataset as part of this project.

The proposed architecture naturally extends to the recognition of expressive playing techniques (e.g., hammer-ons, pull-offs). This can be achieved by adding an auxiliary output head—a $6 \times K$ matrix encoding technique probabilities per string—trained with standard classification losses.

4 Experimental Plan

To validate the proposed multimodal approach, we will conduct a series of controlled experiments comparing audio-only and audio-visual variants of our architecture. The experimental pipeline consists of three core components:

4.1 Dataset Curation

We will record a synchronized audio-video dataset of solo guitar performances specifically designed for tablature transcription evaluation. The dataset will include:

- Diverse playing techniques (hammer-ons, pull-offs, slides, harmonics)
- Close-up video of the fretboard and soundhole region for explicit string visibility
- Annotations of string activations and fret positions

While modest in scale, this dataset will address the critical gap in publicly available multimodal guitar transcription resources.

4.2 Evaluation Metrics

Beyond conventional frame-level pitch metrics (note-wise F1), we will adapt metrics specifically targeting tablature quality, emphasizing correct onset detection. These metrics will reflect the practical utility of transcriptions for guitarists, prioritizing fingering correctness over purely pitch-based evaluation.

4.3 Comparative Evaluation

We will train and evaluate several model variants:

- 1 Audio-only baseline (spectrogram input only)

2 Full multimodal fusion (proposed architecture)

All variants will share identical backbone architectures to isolate the contribution of modality fusion. We hypothesize that the multimodal model will demonstrate statistically significant improvement in string-fret accuracy while the audio-only baseline will maintain superior pitch estimation in visually ambiguous scenarios (e.g., heavy hand occlusion). This controlled comparison will quantify the complementary value of visual information for resolving the fundamental string-pitch ambiguity inherent to guitar transcription.

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